



Production Readiness Assessment

What must happen to take the prototype to a deployed, secure, monitored production MVP — prepared for the board approving the deployment budget.

Document	3 of 3 — Production Readiness Assessment
Date	June 10, 2026
Basis	Build telemetry through commit 7af2303
MVP scope	4 capability areas (see §2)
Overall rating	2.2 / 5.0 — not yet production-ready
Recommendation	Approve build-out; gate go-live on §10 criteria

CONFIDENTIAL — Prepared for internal planning & deployment approval. Estimates are rough-order-of-magnitude planning figures, not commitments.

1. Executive summary

The prototype proves the product thesis end to end, but it runs today as local developer processes against a single database with demo authentication. It is not yet a production system. This assessment sets out, at board level, exactly what must be built and hardened to deploy a secure, monitored MVP: cloud infrastructure, security and FERPA compliance, CI/CD, monitoring, the decomposition into production services, and DNS/networking. None of these require re-architecting the prototype — the event-driven design and service boundaries are sound — they require productionizing it.

We recommend approving the build-out. The architecture is validated and the path is well-understood; the residual risk is execution scope (a lean team) and compliance timing, both addressed in the plan. Go-live should be gated on the criteria in §10.

2. MVP scope & what is proven

MVP capability	Scope	Status	Production gap
1 • Teacher Onboarding & Setup	Sign-in, roster sync, classroom context, initial setup	PARTIAL	Real IAM/SSO, live Clever/ClassLink sync + dedup, MFA, setup flows
2 • Planning & Orchestration	Identify needs, surface next-best actions, suggest schedule	NET-NEW	New orchestration service: needs engine + scheduling over progress data
3 • Assignment & Progress	Assign activities, view student progress, connect data to platform	VALIDATED	Harden & scale the proven teacher loop; broaden activity types
4 • Admin & Leadership View	Usage visibility, operational dashboard, reporting	PARTIAL	New reporting/analytics service + usage telemetry + dashboards

Proven in the prototype (de-risked):

- End-to-end event-driven loop: an offline assessment session syncs, flows over Kafka via a transactional outbox, and a separate service advances curriculum to mastered — verified in-browser.
- At-least-once durability: events survived a consumer outage and drained on restart — no data loss.
- Clean service boundaries (no cross-service DB access) with per-consumer CQRS read models.
- Offline-first client (IndexedDB outbox + cursor delta-sync) confirmed in a real browser.
- Integrated predictive model: a scikit-learn logistic regression (hold-out accuracy 0.74, ROC-AUC 0.81) scores IEP goal attainment and is surfaced as color-coded risk on student records plus a leadership roll-up — served by a dedicated Python service over the shared data spine.

3. Readiness scorecard

Domain	Status	Score	Summary
Architecture & service design	GREEN	4.5	Sound, validated event-driven design; productionize, don't redesign
Assignment & Progress (Cap 3)	GREEN	4.0	Proven teacher loop; needs hardening & scale
Predictive model (goal risk)	AMBER	3.0	Built & integrated (logreg, risk-banded); needs retraining pipeline, drift monitoring & governance
Cloud infrastructure	RED	1.5	Local processes + single DB; no LB, auto-scaling, cache, or CDN
Security (encryption/authz/secrets)	RED	1.0	Demo cookie auth; secrets in .env; no at-rest encryption config
FERPA / student-data compliance	RED	1.0	No DPAs, retention, DSAR, audit logging, or SOC 2 — top blocker
CI/CD & release process	RED	1.5	Clean builds; no pipeline, GitOps, or release/rollback process
Monitoring & observability	RED	1.5	App logs only; no APM, alerting, error tracking, or uptime checks
Onboarding & identity (Cap 1)	AMBER	2.0	Roster graph + admin onboarding exist; no real SSO or live sync
Admin & Leadership view (Cap 4)	AMBER	2.0	Shell + data exist; reporting service & dashboards net-new
Planning & Orchestration (Cap 2)	RED	1.0	Net-new; data foundation present
DNS & networking	RED	1.0	localhost only; no domain, managed TLS, or CDN
Accessibility (WCAG 2.1 AA)	RED	1.0	Not audited — legal & mission requirement for the audience

Overall: 2.3 / 5.0 — a validated MVP, not yet production-ready. Green ≥ 3.5 · Amber 2.0–3.4 · Red < 2.0 .

4. Cloud infrastructure

Element	Current (prototype)	Production target
Compute	Local Node processes; `next start`	Containerized services on Amazon EKS (Fargate/EC2), rolling deploys
Load balancing	None — direct ports	AWS ALB with health checks; TLS termination; path/host routing
Auto-scaling	None	Horizontal Pod Autoscaler (CPU/latency) + cluster autoscaler; queue-depth scaling for workers
Database	Single Postgres, co-located schemas	Aurora PostgreSQL, one cluster per service, multi-AZ + read replicas
Caching	None	ElastiCache (Redis): session, read-through read-models, rate-limiting
CDN	None	CloudFront (US edge) for web/static assets and signed media

Recommendation: stand up the EKS + ALB + Aurora + Redis + CloudFront baseline via Terraform in WS-A (M1–M2); it underpins every capability and all later hardening.

5. Security

Element	Current (prototype)	Production target
Encryption	TLS in dev only; no at-rest config	KMS at-rest (Aurora, S3, EBS, backups); TLS 1.2+ in transit; field-level for sensitive PII
Auth / authz	Demo base64 cookie + Cedar policies	Cognito SSO (Clever/ClassLink, SAML, LTI 1.3) + MFA for staff; short-lived signed sessions; Cedar retained
Secrets management	.env files on disk	AWS Secrets Manager with rotation; no secrets in code or CI logs
FERPA compliance	Data minimization started	Full controls program — see callout below

FERPA / student-data compliance (gating). As a "school official" handling education records under FERPA, the platform must implement: signed Data Processing Agreements per district; a data map and inventory; least-privilege access with full audit logging; configurable data retention and deletion; parental/eligible-student access & correction (DSAR) workflows; sub-processor governance; breach detection & notification; and COPPA handling for under-13 students. SOC 2 readiness provides the control evidence. This is the single largest go-live blocker and must start in week 1.

6. CI/CD

Element	Current (prototype)	Production target
Source control	Git / GitHub; clean builds, lint, types	GitHub with branch protection, CODEOWNERS, required checks, signed commits
Deployment pipeline	Manual local builds	GitHub Actions → ECR → Argo CD (GitOps); Terraform IaC; ephemeral preview envs

Element	Current (prototype)	Production target
Release process	None	Promote dev → staging → prod; canary / blue-green; automated rollback; gated DB migrations
Cache invalidation	Manual; service-worker cache risk	Content-hashed assets + CloudFront invalidation on deploy + service-worker version bump + explicit cache-control

Note: the PWA already ships a service worker; its cache must be versioned and busted on each release to avoid serving stale offline assets — handled in the release pipeline.

7. Monitoring

Element	Current (prototype)	Production target
APM	None	Distributed tracing + per-service latency/throughput (OpenTelemetry → Datadog/Grafana)
Logging	Console logs	Centralized structured logs (CloudWatch/OpenSearch) with retention & PII scrubbing
Alerting	None	SLO-based alerts → PagerDuty on-call; sync-freshness & dead-letter-queue monitors
Error tracking	None	Sentry across web and services with release tagging
Uptime	None	Synthetic checks (CloudWatch Synthetics / Pingdom) + public status page

8. Services architecture — prototype → production

The prototype already separates concerns cleanly; production decomposes the co-located dev setup into independently deployable, scalable services on the shared event backbone.

Prototype component	Production service(s)	Production change
Single Postgres (co-located schemas)	Per-service Aurora clusters	DB-per-service split; multi-AZ; read replicas
roster-graph service	Roster Graph + IAM/Identity + Roster Sync	Add Cognito + Clever/ClassLink sync workers + dedup
student-record service	Student Record & Outcomes	Scale outbox relay (HA) or Debezium CDC
soler service	Assessment (SOLER) + offline-sync gateway	Scale; harden delta-sync endpoints
links service	Curriculum (Links) + projector workers	Independent consumer scaling
(net-new)	Planning & Orchestration service	Next-best-action (rules) + scheduling over progress data
(net-new)	Reporting & Analytics service	Usage telemetry + leadership dashboards + exports
In-memory / Kafka broker	Amazon MSK + Schema Registry	Already validated on Kafka; add registry + DLQ
Single-instance outbox relay	Relay workers (SKIP LOCKED) or CDC	Remove single point of failure
Next.js web (`next start`)	Web app on EKS behind ALB + CloudFront	Containerized, autoscaled, CDN-fronted
Python prediction service (uvicorn)	Prediction service (containerized) + retraining job	Scale horizontally; scheduled retraining; model registry

Predictive model — path to production (MLOps)

Element	Current (prototype)	Production target
Model serving	FastAPI loads a joblib artifact	Containerized service (autoscaled) loading from a versioned model registry (S3/MLflow)
Feature pipeline	Live reads from the co-located DB	Service-owned read model fed by the student.metric.v1 event backbone; reproducible feature set
Retraining	Manual `train.py`	Scheduled retraining + offline evaluation gate before promotion; reproducible runs
Monitoring	None	Prediction & data drift, calibration, and performance dashboards with alerts
Governance	Model card (metrics + coefficients)	Bias/fairness review across student subgroups; human-in-the-loop; FERPA-aware audit of inferences

Recommendation: the model is integrated and useful today; treat it as *decision support* (a teacher/leadership flag), keep a human in the loop, and add the MLOps controls above before it influences any high-stakes action.

9. DNS & networking

Element	Current (prototype)	Production target
Domain config	localhost	Route 53 hosted zone; app + api subdomains; health-checked records
SSL / TLS	Dev only	ACM-managed certs; TLS 1.2+; HSTS; auto-renewal
US content delivery	None	CloudFront US-only edge (geo-restriction) over US AWS regions; data residency in-US
Edge protection	None	AWS WAF + Shield (DDoS); rate-limiting; bot controls at the edge

10. What has to happen before go-live (gates)

Pri	Gate	Why it gates go-live
P0	Production IAM + SSO + MFA	Replaces demo auth; gates all real access
P0	FERPA program + signed DPAs + audit logging	Legal prerequisite to handle student records
P0	Secrets management + at-rest encryption (KMS)	Baseline data protection
P0	Cloud baseline: EKS+ALB+Aurora+Redis+CloudFront	Deployable, scalable footprint
P0	CI/CD pipeline with rollback + monitoring/alerting	Safe to deploy and operate
P0	Penetration test passed + WCAG 2.1 AA audit	Security + accessibility sign-off
P1	Live roster connectors + identity dedup	Real onboarding at scale
P1	Load test to initial scale + DR/backup drill	Reliability evidence
P1	Planning/Orchestration + Admin dashboards live	Complete the MVP surface
P2	Model MLOps: retraining, drift monitoring, bias review	Keep predictions trustworthy & governed over time

Assessment basis: live verification of the running services (PostgreSQL + Kafka), an in-browser walkthrough of the offline teacher loop, and review of the source, ADRs and architecture docs as of Build telemetry through commit 7af2303.